Waqas Rabbani

SPMM COURSE

Started on	Sunday, 24 July 2022, 5:40 PM
State	Finished
Completed on	Sunday, 24 July 2022, 5:40 PM
Time taken	8 secs
Marks	0.00/195.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 8.00

Flag question

Blood supply to brain

Match the specific areas of the brain with their blood supply from the list provided

Anterior cerebral artery — Choose...

Middle cerebral artery — Choose...

Posterior cerebral artery — Choose...

Vertebral arteries — Choose...

Basilar artery — Choose...

Choose...
Internal capsule
Inferomedial temporal lobe, Occipital lobe
Medial aspects of cerebral cortex, Superior lateral aspects of cerebral cortex
Pons
Medulla
Lateral aspects of cerebral cortex, Broca's & Wernicke's area

Explanation:

The internal carotid artery enters the circle of Willis and divides to form the anterior cerebral and middle cerebral arteries. The anterior cerebral artery supplies the medical and superior lateral aspects of the cerebral cortex to the parietal/occipital border.

The middle cerebral artery supplies the greater part of the lateral aspect of the cerebral cortex, including the Broca's and Wernicke's area in the dominant hemispheres.

The posterior cerebral artery arises from basilar artery and supplies the inferomedial temporal lobe and the occipital lobe. The internal capsule is supplied by branches from the circle of Willis.

The posterior inferior cerebellar artery and anterior spinal artery, branches of the vertebral artery, supply medulla.

The basilar artery that runs along the midline of the pons supplies Pons.

The correct answer is: Anterior cerebral artery → Medial aspects of cerebral cortex, Superior lateral aspects of cerebral cortex, Middle cerebral artery → Lateral aspects of cerebral cortex, Broca's & Wernicke's area, Posterior cerebral artery → Inferomedial temporal lobe, Occipital lobe, Vertebral arteries → Medulla, Basilar artery → Pons

Question **2**

Not answered

Marked out of 4.00

Cerebral arteries & lesions

Using the clinical descriptions below, identify the artery most likely to be involved

A 48-year-old gentleman presenting in an akinetic mute state has quadriparesis with legs more affected than the arms.

A 64-year-old gentleman develops brain stem ischemia during exercise. When arms are held straight, there is a drop in more than 30mm Hg between the arms

A 57-year-old gentleman who is a right handed individual presenting with left hemiparesis and sensory loss involving face & arms. He also neglects his right side

A 67-year-old gentleman with presenting with left sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared.

Choose...

Choose...
Right middle cerebral artery
Posterior inferior cerebellar artery
Left middle cerebral artery
Bilateral anterior cerebral artery
Right anterior cerebral artery
Basilar artery
Posterior cerebral artery
Left anterior cerebral artery
Subclavian artery proximal to ipsilateral vertebral artery
Vertebral artery

Explanation:

The effects of bilateral anterior cerebral artery infarct include quadriparesis with legs more affected than arms and akinetic mutism (ventromedial or cingulate syndrome). The lesion, in this case, would be in the cingulate gyri. The anterior cerebral artery supplies the medial surface of the brain, including the ventromedial frontal lobe, the cingulum, the premotor cortex, and the motor strip.

Subclavian steel syndrome is caused due to lesions involving subclavian artery proximal to the ipsilateral vertebral artery. The patients may present with brain stem ischaemia following arm exertion, causing a difference in BP of more than 20 mm Hg between the arms.

Middle cerebral artery lesions are characterized by hemiplegia, hemianesthesia. If the lesion involves non-dominant hemisphere, then contralateral neglect is highly likely.

Anterior cerebral artery lesions involve contralateral hemiplegia/hemiparesis with weakness involving more of the legs than the arms, and the face will usually be spared.

Excerpt retrieved from Frontal Lobe Syndromes Clinical Presentation - Medscape Reference. (n.d.). Retrieved from http://emedicine.medscape.com/article/1135866-clinical

The correct answer is: A 48-year-old gentleman presenting in an akinetic mute state has quadriparesis with legs more affected than the arms. → Bilateral anterior cerebral artery, A 64-year-old gentleman develops brain stem ischemia during exercise. When arms are held straight, there is a drop in more than 30mm Hg between the arms → Subclavian artery proximal to ipsilateral vertebral artery, A 57-year-old gentleman who is a right handed individual presenting with left hemiparesis and sensory loss involving face & arms. He also neglects his right side → Right middle cerebral artery, A 67-year-old gentleman with presenting with left sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared. → Right anterior cerebral artery

Question **3**

Not answered

Marked out of 4.00

Brain lesions

For each of the following clinical presentation, identify the most likely brain lesion

A 77-year-old man has suffered a head injury and is now incontinent. He has limited verbal output and lacks spontaneous movement — Choose...

A 63-year-old man suffered a car accident and became quadriplegic. He is conscious but not able to speak or vocalise. He can respond to questions by blinking — Choose...

A 69-year-old man has marked difficulties with seeing objects properly. The CT trainee holds a pen, and he reports 'I see a pen.....I think'. Then the trainee holds up a mobile phone and the man reports 'I see a phone'. When the trainee now holds the pen and mobile phone overlapping one another he says, I think I see a pen. When the trainee asks again, the man says 'Wait.....I see a phone.' — Choose...

A 56-year-old man has suffered a head injury and presented with memory difficulties. He is apathetic, planning deficits and impairment of working memory. — Choose...

Choose...
Bilateral calcarine lesion
Optic tract lesion
Non dominant temporal lobe lesion
Dominant temporal lobe lesion
Medial frontal lesion
Dorsolateral prefrontal cortex lesion
Dominant parietal lobe lesion
Bilateral posterior parietal lesion
Non-dominant parietal lobe lesion
Pontine lesion
Cervical spine lesion

Explanation: Medial frontal syndrome (ventromedial), also called as apathetic or pseudo depressive type is featured by a paucity of spontaneous behaviours, sparse verbal output and reduced social knowledge. Incontinence is also a feature of frontal lobe impairment.

In severe lesions involving the pons, patients can present with weakness of arms & legs with dysphagia and loss of speech. However, they can respond to questions by blinking.

Case 3: This patient could be suffering from simultanagnosia (inability to attend to more than one item of a complex scene at a time), a feature of Balint's syndrome resulting from lesions of the bilateral posterior parietal regions. Balint's syndrome consists of a triad of simultanagnosia, optic ataxia (inability to guide reaching or pointing despite adequate vision), and oculomotor apraxia (inability to voluntarily direct saccades to a visual target). Visual fields are unaffected, and oculocephalic reflexes are intact.

Dorsolateral pre-frontal cortex dysfunction is called as dysexecutive or disorganized type. It is featured by executive dysfunction, diminished judgment, and planning and poor insight. The patients are also described to be concrete and inflexible in nature.

The correct answer is: A 77-year-old man has suffered a head injury and is now incontinent. He has limited verbal output and lacks spontaneous movement → Medial frontal lesion, A 63-year-old man suffered a car accident and became quadriplegic. He is conscious but not able to speak or vocalise. He can respond to questions by blinking → Pontine lesion, A 69-year-old man has marked difficulties with seeing objects properly. The CT trainee holds a pen, and he reports 'I see a pen.....I think'. Then the trainee holds up a mobile phone and the man reports 'I see a phone'. When the trainee now holds the pen and mobile phone overlapping one another he says, I think I see a pen. When the trainee asks again, the man says 'Wait.....I see a phone.' → Bilateral posterior parietal lesion, A 56-year-old man has suffered a head injury and presented with memory difficulties. He is apathetic, planning deficits and impairment of working memory. → Dorsolateral prefrontal cortex lesion

Question **4**

Not answered

Neuroanatomical localization (1)

Identify the neuroanatomical location of lesions for the clinical presentations described below:

Marked out of 5.00

Choose...
Dorsal occipital area
Bilateral anterior parietal area
Lateral temporal lobe
Posterior superior temporal lobe
Cerebellum
Medial temporal lobe
Occipito-temporal area
Basal forebrain
Supplementary motor area
Lateral prefrontal cortex

A 57-year-old woman is unable to recognize faces — Choose...

An 83-year-old woman has difficulty in learning new information — Choose...

A 57-year-old man is unable to understand speech but appears fluent. — Choose...

A 49-year-old man can see clearly in one small part of the visual field in the centre while other areas appear foggy since he sustained a head injury. — Choose...

A 58-year-old man in a brain injury unit is described to have an expressionless face. He cannot communicate and only has tracking eye movements preserved. — Choose...

Explanation:

The area of the brain involved in recognizing faces is fusiform gyrus, which is situated in the occipitotemporal region. This patient is unable to learn new information, which is anterograde amnesia, often seen in patients with Alzheimer's disease. The pathology is mainly in a medial temporal lobe, which lodges the hippocampus.

Wernicke's area is situated in the posterior superior temporal lobe and is involved in language comprehension.

Case 4 illustrates key-hole vision that is a feature of cortical blindness involving the dorsal occipital area of the brain.

Anterior cerebral artery supplies most of the medial surface of the brain (structures include ventromedial frontal lobe, the cingulum, the premotor cortex, supplementary motor area and the motor strip). In lesions involving supplementary motor areas, the patients can present with akinetic mutism with only tracking movements of the eyes being preserved.

The correct answer is: A 57-year-old woman is unable to recognize faces → Occipito-temporal area, An 83-year-old woman has difficulty in learning new information → Medial temporal lobe, A 57-year-old man is unable to understand speech but appears fluent. → Posterior superior temporal lobe, A 49-year-old man can see clearly in one small part of the visual field in the centre while other areas appear foggy since he sustained a head injury. → Dorsal occipital area, A 58-year-old man in a brain injury unit is described to have an expressionless face. He cannot communicate and only has tracking eye movements preserved. → Supplementary motor area

Question **5**

Not answered

Marked out of 8.00

Connecting bundles

Identify the areas of the brain that are connected with each other

Choose...
Wernicke's area
Inferior frontal gyrus
Parietal
Dorsomedial thalamus
Lateral geniculate body
Inferior colliculi
Medial frontal gyrus
Superior colliculi
Medial geniculate body
Mamillary bodies

Uncinate fasciculus connects temporal lobe to which of the listed structures? — Choose...

Which of the listed options refers to the thalamic nuclei of the auditory system connected to midbrain? — Choose...

Arcuate fasciculus connects Broca's area with which of the listed structures? — Choose...

Which of the listed options refers to the midbrain structure of visual tract connected to the thalamus? — Choose...

Explanation:

The uncinate fasciculus is a white matter tract that connects parts of the limbic system such as hippocampus and amygdala in the temporal lobe with the orbitofrontal region. It mainly connects the inferior frontal gyrus with temporal lobe.

Mediate geniculate body and inferior colliculi are two brain stem nuclei of the auditory system. Note that the lateral geniculate body and superior colliculi are involved in visual processing.

The arcuate fasciculus is the tract that connects the posterior part of the temper-parietal junction with the frontal cortex in the brain. It mainly connects Wernicke's and Broca's areas.

Lateral geniculate body and superior colliculi are two brain stem nuclei of the visual system.

The correct answer is: Uncinate fasciculus connects temporal lobe to which of the listed structures? → Inferior frontal gyrus, Which of the listed options refers to the thalamic nuclei of the auditory system connected to midbrain? → Medial geniculate body, Arcuate fasciculus connects Broca's area with which of the listed structures? → Wernicke's area, Which of the listed options refers to the midbrain structure of visual tract connected to the thalamus? → Superior colliculi

Question **6**

Not answered

Marked out of 11.00

Developmental organization

For each of the vesicles below, identify the brain structures that develop from it

Telencephalon Choose...

Diencephalon Choose...

Mesencephalon Choose...

Metencephalon Choose...

Myelencephalon Choose...

Choose...
Adrenal medulla
Cerebellum
Caudal part of the medulla oblongata
Thalamus
Vagus nerve
Corpus striatum
Tectum and Tegmentum

Explanation: During embryonic development the midline neural tube differentiates into the following vesicles.

- Prosencephalon which differentiates into the Telencephalon (cerebrum, striatum and pallidum) and Diencephalon (thalamus, subthalamus, pineal body, hypothalamus and epithalamus).
- Mesencephalon (midbrain)has two parts - tectum or corpora quadrigeminal, made up of the superior & inferior colliculi and tegmentum containing the red nucleus and periaqueductal grey matter.
- Rhombencephalon which differentiates into Metencephalon that forms pons, the rostral part of the medulla oblongata and cerebellum and Myelencephalon that forms the caudal part of medulla oblongata.

The correct answer is: Telencephalon → Corpus striatum, Diencephalon → Thalamus, Mesencephalon → Tectum and Tegmentum, Metencephalon → Cerebellum, Myelencephalon → Caudal part of the medulla oblongata

Question 7

Choose...
Nucleus Accumbens
Angular gyrus
Superior parietal cortex
Medulla
Fusiform gyrus
Left inferior frontal area
Cerebellar vermis
Hippocampus
Inferior olivary nucleus
Amygdala

Neuroanatomical regions (1)
Identify one brain area that is most important in the following functions;

Reward processing	Choose...
Prosopagnosia	Choose...
Recalling previously learned material	Choose...

Explanation:

The brain regions implicated in reward system regulation include nucleus accumbens and ventral tegmental area.

The area of the brain involved in mediating emotional response such as fear is an amygdala. Bilateral amygdalar damage is associated with a loss of fear conditioning, as evidenced from the comprehensive investigations in patient SM who has complete, focal bilateral amygdala damage.

The brain area involved in speech production is Broca's area, which is situated, in the left inferior frontal cortex.

Prosopagnosia is difficulty in recognizing faces, mediated through fusiform gyrus.

Hippocampus is involved in recalling previously learned material

Excerpt retrieved from Tranel et al. (2006), Altered experience of emotion following bilateral amygdala damage. http://www.ncbi.nlm.nih.gov/pubmed/17354069

The correct answer is: Reward processing → Nucleus Accumbens, Prosopagnosia → Fusiform gyrus, Recalling previously learned material → Hippocampus

Question 8

Not answered

Choose...
Basal ganglia
Posterior superior temporal lobe
Hypothalamus
Cerebellum
Corpus callosum
Medulla
Left inferior frontal area
Anterior cingulate cortex
Amygdala
Dorsolateral prefrontal cortex
Hippocampus

Neuroanatomical regions (2)
Identify one brain area that is most important in each of the following functions

Planning and programming of movement	Choose...
Motivational circuitry	Choose...
Preparing a motor plan and predicting balance	Choose...
Executive functions	Choose...
Comprehension of speech	Choose...

Explanation: The basal ganglia are involved in the planning and programming of movement. They also have a role in the processes by which an abstract thought is converted into voluntary action.

The motivation circuitry involves anterior cingulate cortex and its reciprocal connections with the rest of the prefrontal cortex.

The cerebellum has the important role of preparing a motor plan and predicting balance needed between muscle groups to carry out the intended action smoothly. Cerebellar lesions produce ataxia and coarse intentional tremors, along with hypotonia, past pointing and pendular knee jerk.

The dorsolateral prefrontal cortex is responsible for executive functions such as planning, judgement, and decision-making.

Wernicke's area, which is situated in the posterior superior temporal lobe, is involved in speech comprehension.

The correct answer is: Planning and programming of movement → Basal ganglia, Motivational circuitry → Anterior cingulate cortex, Preparing a motor plan and predicting balance → Cerebellum, Executive functions → Dorsolateral prefrontal cortex, Comprehension of speech → Posterior superior temporal lobe

Question **9**

Not answered

Marked out of 4.00

Neuroanatomical localization (2)

Identify the brain regions where the following structures are situated

Entorhinal cortex	Choose...
Striatum	Choose...
Ventral tegmental area	Choose...
Hypothalamus	Choose...

Choose...
Temporal lobe
Diencephalon
Occipital lobe
Parietal lobe
Pons
Basal ganglia
Prefrontal cortex
Medulla oblongata
Cerebellum
Midbrain

Explanation:

The entorhinal cortex is situated in the temporal lobe.

The basal ganglia are a group of gray matter nuclei that forms the largest subcortical structure in the brain. The components of the basal ganglia include the caudate nucleus, the putamen and the globus pallidus. Putamen and globus pallidus together forms the lentiform nucleus. Caudate and putamen are often clubbed together by the name striatum.

The ventral tegmental area is situated in the midbrain.

The hypothalamus is a part of the diencephalon.

The correct answer is: Entorhinal cortex → Temporal lobe, Striatum → Basal ganglia, Ventral tegmental area → Midbrain, Hypothalamus → Diencephalon

Question **10**

Not answered

Marked out of 4.00

Pathophysiology of Schizophrenia

Identify the one important area of the brain involved in each of the following pathophysiological findings in schizophrenia

D2 occupancy in this region by antipsychotics, predicts antipsychotic response	Choose...
Enlargement of this structure can be seen as a consequence of long-term treatment with haloperidol	Choose...
A consistent reversal of normal left larger than right asymmetry of this structure is noted	Choose...
Neuroimaging studies suggest decreased activity in this part of the brain	Choose...

Choose...
Lateral ventricles
Prefrontal cortex
Corpus striatum
Amygdalar volume
Heschl's gyrus
Dorsal raphe
Occipital pole
Thalamic volume
Left temporal horn
Brain weight & volume
Planum temporale

Explanation: Dopamine receptor occupancy (D-2) in the corpus striatum predicts response to antipsychotics. It also plays an important role in causing akathisia and other extrapyramidal side effects caused by antipsychotics.

Enlargement of basal ganglia can be seen in schizophrenia as a consequence of treatment with classical neuroleptics, which can be reversed by the use of atypical substances.

The planum temporale, the posterior superior surface of the superior temporal gyrus, is a highly lateralized brain structure involved with language. In schizophrenic patients, a consistent reversal of the usual left-larger-than- right asymmetry of planum temporale surface area is noted.

Neuroimaging studies suggest decreased activity in the prefrontal cortex of the brain, relating to negative symptoms of schizophrenia and reduced motivation.

The correct answer is: D2 occupancy in this region by antipsychotics, predicts antipsychotic response → Corpus striatum, Enlargement of this structure can be seen as a consequence of long-term treatment with haloperidol → Corpus striatum, A consistent reversal of normal left larger than right asymmetry of this structure is noted → Planum temporale, Neuroimaging studies suggest decreased activity in this part of the brain → Prefrontal cortex

Question **11**

Not answered

Marked out of 9.00

Personality
Constructional apraxia
Memory
Aphasia
Neglect
Face Recognition
Visuospatial deficits
Dyscalculia
Limb apraxia

HiY Neurosciences CXEMI061

Neuroanatomical localization (3)

Identify 3 important functions for each of the following;

1. Dominant hemisphere

2. Non-dominant hemisphere

3. Bilateral and not strictly localised

Explanation:

Dominant hemisphere functions include aphasia (language), calculation, praxis (limb apraxia).

Non-dominant hemisphere function includes visuospatial deficits, constructional apraxia, dressing apraxia, neglect and impaired visual perception.

Face recognition involves both hemispheres.

Broader psychological constructs such as personality and memory are not strictly localised to brain regions.

The correct answer is:

Dominant hemisphere (Choose THREE)- Aphasia, Dyscalculia, Limb apraxia,

Non-dominant hemisphere (Choose THREE)- Constructional apraxia, Neglect, Visuospatial deficits

Bilateral and not strictly localised (Choose THREE)- Personality, Face Recognition, Memory

Question **12**

Not answered

Marked out of 4.00

Foramen of Magendie
Foramen of Lucas
Jugular foramen
Foramen of Luschka
Foramen Rotandum
Foramina of Monroe
Foramen ovale
Internal auditory meatus
Cerebral aqueduct of Sylvius
Foramen magnum

HiY Neurosciences CXEMI062

CSF Flow

Identify the connections that facilitate the flow of CSF

1. Lateral ventricles to 3rd ventricles

2. 3rd ventricles to 4th Ventricles

3. 4th Ventricle to subarachnoid space (2 answers)

CSF is secreted by the choroid plexus in the lateral, third and fourth ventricles and at a rate of 300 ml/day, which is almost protein free. Route: From the lateral ventricle to 3rd ventricle via interventricular foramina of Monroe. From 3rd to 4th Ventricle via cerebral aqueduct of Sylvius. From 4th ventricle to subarachnoid space via Foramen of Magendie (middle) and Foramen of Luschka (two, lateral).

Question **13**

Not answered

Marked out of 5.00

Choose...
Simultanagnosia
Prosopagnosia
Anton syndrome
Horner syndrome
Achromatopsia
Alexia without agraphia
Korsakoff syndrome
Finger Agnosia
Marchiafava- Bignami syndrome
Kluver-Bucy syndrome

Brain lesions & syndromes

For each of the clinical descriptions given below, identify the terms used to describe them

A 57-year-old man with history of blindness following major stroke denies any visual deficits and attempts to walk in his room without any aid — Choose...

This is one of the primary symptoms of Gerstmann syndrome — Choose...

A 67-year-old woman has suffered from a watershed infarction of the brain. Following this event, she is unable to attend to more than one item of a complex scene at a time — Choose...

A 77-year-old man with history of recent cerebral infarct is unable to discriminate colours — Choose...

A 67-year-old woman can normally speak and comprehend what is spoken. She can write spontaneously, but reading comprehension is impaired. — Choose...

Explanation:

Anton syndrome is a visual agnosia, in which the patient denies any deficit and may attempt to negotiate the environment, invariably without success. In the curious phenomenon known as blindsight, visual stimuli can induce a response despite cortical blindness.

Gerstmann syndrome is characterized by four symptoms: dysgraphia/agraphia, dyscalculia/acalculia, finger agnosia and right-left disorientation. It is rarely seen as a full tetrad. It is often associated with

brain lesions in the dominant parietal lobe (angular and supramarginal gyri).

Balint syndrome is a triad of simultanagnosia (inability to attend to more than one item of a visual scene at the same time), optic ataxia (inability to guide one's reach despite intact vision), and oculomotor apraxia (inability to freely direct visual saccades to a target). Possible causes include carbon monoxide poisoning, watershed infarction, leukodystrophy, and the posterior cortical variant of Alzheimer's disease.

Colour processing deficits such as achromatopsia (loss of ability to discriminate colours) are often associated with pure alexia after medial occipitotemporal damage, following left posterior cerebral artery infarction (Kipps & Hodges, 2005).

Pure word blindness (alexia no agraphia): Here the patient can normally speak and comprehend what is spoken; he can also write spontaneously and to dictation, but reading comprehension is impaired.

Kipps and Hodges (2005), Cognitive assessment for clinicians. Excerpt retrieved from http://jnnp.bmj.com/content/76/suppl_1/i22.full

The correct answer is: A 57-year-old man with history of blindness following major stroke denies any visual deficits and attempts to walk in his room without any aid → Anton syndrome, This is one of the primary symptoms of Gerstmann syndrome → Finger Agnosia, A 67-year-old woman has suffered from a watershed infarction of the brain. Following this event, she is unable to attend to more than one item of a complex scene at a time → Simultanagnosia, A 77-year-old man with history of recent cerebral infarct is unable to discriminate colours → Achromatopsia, A 67-year-old woman can normally speak and comprehend what is spoken. She can write spontaneously, but reading comprehension is impaired. → Alexia without agraphia

Question **14**

Not answered

Marked out of 7.00

Neurohistology

Match the following neuroglial cells with their specific functions from the list provided:

Astrocytes	Choose...
Oligodendrocytes	Choose...
Microglia	Choose...
Ependymal cells	Choose...
Schwann cells	Choose...

Choose...
Peripheral myelin sheath formation
Breakdown of some neurotransmitters
Neuronal support in sensory and autonomic ganglia
Apoptotic clearance
CNS myelin sheath formation.
Aids the flow of CSF
Neurilemma formation
Act as scavenger cells at sites of CNS injury

Explanation: Astrocytes are the most numerous of all three types of glial cells. These are star-shaped cells that enable nutrition of neurons, breakdown of some neurotransmitters, and maintaining the blood-brain barrier.

Oligodendrocytes are seen in CNS (not in peripheral nerves, where Schwann cells replace them). They produce myelin sheaths that help in the saltatory conduction of an action potential (pole to pole jumping), which quicken the process of signal transmission.

The microglia are descendants of macrophages. They are scavenger cells that clear neuronal debris following cell death.

Ependymal cells are a particular type of glia that cover the ventricles and enable CSF circulation.

Schwann cells are seen in the peripheral nervous system where they produce myelin sheaths.

The correct answer is: Astrocytes → Breakdown of some neurotransmitters, Oligodendrocytes → CNS

myelin sheath formation., Microglia → Act as scavenger cells at sites of CNS injury, Ependymal cells → Aids the flow of CSF, Schwann cells → Peripheral myelin sheath formation

Question **15**

Not answered

Marked out of 6.00

Neuroanatomical lesions in CNS disorders

For each of the condition below choose the most specific option/s from given list

Korsakoff's psychosis Choose...

Gerstmann syndrome Choose...

Prosopagnosia Choose...

Alzheimer's disease Choose...

Choose...
Inferior frontal gyrus
Medial Temporal Lobe
Corpus callosum
Hypothalamus
Mamillary bodies
Dorsolateral prefrontal cortex
Angular gyrus
Fusiform gyrus
Third ventricle

Explanation:

About 80 per cent of alcoholic patients recovering from Wernicke's encephalopathy develop Korsakoff's amnesic syndrome. Diagnosis of Korsakoff's syndrome correlates best with the presence of lesions in the dorsomedial thalamus. But lesions in the mammillary bodies, the mammillothalamic tract and the anterior thalamus may be more relevant to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus (Kopelamn et al., 2009).

Gerstmann syndrome is characterized by four primary symptoms: Dysgraphia/agraphia, Dyscalculia/acalculia, Finger agnosia and Left-right disorientation. This is often associated with brain lesions in the dominant (usually left) side of the angular and supramarginal gyri (parietal lobe).

Prosopagnosics cannot recognise familiar faces. Face processing is a bilateral function; more vital areas may be present on the right hemisphere. Acquired prosopagnosia is usually associated with bilateral or right-sided lesions of the occipital - temporal junction (fusiform gyrus).

In Alzheimer's disease, the main areas affected are the hippocampus and medial temporal lobe. Atrophic changes are mainly observed in these two areas in even in very early stages of Alzheimer's disease.

Barton, JJ., (2008). Prosopagnosia associated with a left occipitotemporal lesion. Excerpt retrieved from http://www.ncbi.nlm.nih.gov/pubmed/18374372
Kopeman et al., (2009). The Korsakoff Syndrome: Clinical Aspects, Psychology and Treatment. Excerpt retrieved from http://alcalc.oxfordjournals.org/content/44/2/148

The correct answer is: Korsakoff's psychosis → Mamillary bodies, Gerstmann syndrome → Angular gyrus, Prosopagnosia → Fusiform gyrus, Alzheimer's disease → Medial Temporal Lobe

Question **16**

Not answered

Marked out of

Neurotransmitters

For the following neurotransmitters choose the associated disorders from the given list

Dopamine Choose...

Noradrenaline Choose...

Acetylcholine Choose...

Choose...
Motor neurone disease
Parkinson's disease
Depression
Panic disorder
Obsessive compulsive disorder
Alzheimer's disease
Huntington's disease

GABA Choose...

Explanation: Dopamine is associated with Parkinson's disease and schizophrenia. Noradrenalin is associated with both anxiety and depression. Acetylcholine is associated with Alzheimer's disease. GABA is thought to be associated with schizophrenia, anxiety as well as Huntington's disease.

The correct answer is: Dopamine → Parkinson's disease, Noradrenaline → Depression, Acetylcholine → Alzheimer's disease, GABA → Huntington's disease

Question **17**

Not answered

Marked out of 9.00

Neuropeptides & disorders

Identify the function/disorder associated with each of the neuropeptides listed below

Beta-endorphins Choose...

Vasopressin Choose...

Cholecystokinin Choose...

Orexins Choose...

Met-enkephalin Choose...

Choose...
Mood
Anxiety
Arousal
Circadian function
Cognition
Panic
OCD

Explanation: Endorphins are endogenous opioid neurotransmitters produced by the pituitary and the hypothalamus during exercise, emotional excitement and experience of pain. Endorphins like their synthetic counterparts - opiates, produce analgesia and a feeling of well-being. Beta-endorphin is the most potent natural opioid. It is associated with pain and mood regulation. "Although the effects of vasopressin analogs on cognition are inconsistent, overall (Strupp and Levitsky, 1985), the evolving evidence seems to suggest that vasopressin regulates various aspects of cognition, including learning, memory, and/or attention"(Jentsch, 2003).

Cholecystokinin is associated with anxiety and pain. It has been implicated in the pathophysiologies of schizophrenia, eating disorders and movement disorders. CCK infusion induces panic attacks in people with panic disorder. CCK antagonists may have anxiolytic effects.

Orexins A and B (hypocretins) are closely related neuropeptides derived from a single gene. They act as OX1 and OX2 receptors, which are highly expressed in the lateral hypothalamus and other brain areas involved in stress regulation. Orexins were initially identified as important regulators of feeding but are now seen as involved in circadian function, sleep and response to stress, including neuroendocrine control.

The enkephalins (Leu-enkephalin and met-enkephalin) mediate pain perception, mood and neuroendocrine behaviours; they are found in the brain, spinal cord, nerve plexuses and exocrine glands of the gastrointestinal tract (Ref: J Neurosci. 2003 Feb 1;23(3):1066-71)

The correct answer is: Beta-endorphins → Mood, Vasopressin → Cognition, Cholecystokinin → Anxiety, Orexins → Circadian function, Met-enkephalin → Mood

Question **18**

Not answered

Agnosias & Apraxias (1)

Find one term to describe each of the following presentations

Marked out of 5.00

Choose...
Visuospatial agnosia
Simultagnosia
Astereognosia
Ideomotor apraxia
Hemisomatognosia
Finger agnosia
Dressing apraxia
Anosognosia
Agraphaesthesia
Prosopagnosia

A 76-year-old gentleman is having difficulty recognizing complex pictures — Choose...

An 83-year-old woman is having difficulty copying a cube diagram on extended cognitive examination — Choose...

A 73-year-old gentleman with a history of major stroke has a left sided hemiplegia. However, he denies having any physical problems including the difficulty he has in walking. — Choose...

A 78-year-old gentleman with Alzheimer's dementia has difficulty in recognizing the faces of his children, grandchildren and close friends/ However he recognizes their voices — Choose...

A 76-year-old woman complains of a feeling that the limbs on one side of the body are missing — Choose...

Explanation: **Simultanagnosia** is the inability to recognize complex pictures.

Visuospatial agnosia is the inability to construct or copy figures. Copying three-dimensional shapes such as a wire cube, interlocking pentagons, or constructing a clock-face with numbers are good tests of constructional ability and may also highlight visuospatial neglect if it is present.

Anosognosia is failure to recognize a disability. The patient does not recognise functional deficits of their body.

While patients with **prosopagnosia** cannot recognise familiar faces, they often use other indicators such as voice, clothes or body movements to aid identification (Kipps and Hodges 2005). The identification deficit may not be limited to faces; identifying objects within other categories may also be impaired (for example, different models of motor cars and types of flowers).

Hemisomatognosia is a term used when a part of the body is felt to be absent or neglected.

The correct answer is: A 76-year-old gentleman is having difficulty recognizing complex pictures → Simultagnosia, An 83-year-old woman is having difficulty copying a cube diagram on extended cognitive examination → Visuospatial agnosia, A 73-year-old gentleman with a history of major stroke has a left sided hemiplegia. However, he denies having any physical problems including the difficulty he has in walking. → Anosognosia, A 78-year-old gentleman with Alzheimer's dementia has difficulty in recognizing the faces of his children, grandchildren and close friends/ However he recognizes their voices → Prosopagnosia, A 76-year-old woman complains of a feeling that the limbs on one side of the body are missing → Hemisomatognosia

Question **19**

Not answered

Marked out of 10.00

Choose...
Globus pallidus
Amygdala
Lentiform nucleus
Hippocampus
Orbitofrontal cortex
Caudate nucleus
Corpus striatum
Septal Nuclei
Cingulate gyrus

Constituents of brain regions

Choose the components of the following structures. (Each option can be used once, more than once or not at all)

In addition to putamen, what other structure constitutes the corpus striatum? — Choose...

In addition to putamen, what other structure constitutes the lentiform nucleus? — Choose...

Which frontal structure is a constituent of the Papez circuit? — Choose...

In addition to hippocampus, what other structure constitutes the medial temporal lobe? — Choose...

Explanation:

The basal ganglia are a group of gray matter nuclei that forms the largest subcortical structure in the brain. The components of the basal ganglia include the caudate nucleus, the putamen and the globus pallidus. Putamen and globus pallidus together forms the lentiform nucleus. Caudate and putamen are often clubbed together by the name striatum. Both the subthalamic nuclei and the substantia nigra are physiologically related to the basal ganglia though not considered to be a part of it.

The Papez circuit in its original description consists of the hippocampus, the fornix, the mamillary bodies, the anterior nucleus of the thalamus, and the cingulate gyrus. The boundaries of the limbic system were subsequently expanded to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.

The hippocampus and amygdala are components of the temporal lobe and play important roles within the limbic system.

The correct answer is: In addition to putamen, what other structure constitutes the corpus striatum? → Caudate nucleus, In addition to putamen, what other structure constitutes the lentiform nucleus? → Globus pallidus, Which frontal structure is a constituent of the Papez circuit? → Cingulate gyrus, In addition to hippocampus, what other structure constitutes the medial temporal lobe? → Amygdala

Question **20**

Not answered

Marked out of 4.00

Neuropsychological Tests

For each clinical description below choose ONE appropriate neuropsychological test.

A 55-year-old lady scores a 26/30 on the MMSE. But she responds saying, "I don't know", "I can't be bothered" for many items in the test.

The family of a 60-year-old lady with traumatic brain injury is seeking compensation for the cognitive impairment sustained following RTA. You are asked to provide a report on possible premorbid IQ of this lady.

A 60-year-old man has poor attention span with significant difficulty organizing and planning even simple daytime activities; he vehemently denies any impairment in memory.

A 16-year-old girl is disruptive and aggressive at school. She often loses her pocket money and performs poorly at school. Her brother has a moderate learning disability.

Choose...
Wisconsin Card Sorting Test
Beck Depression Inventory
National Adult Reading Test
Rey-Osterrieth Test
Bender Gestalt Test
Blessed Rating Scale
MMSE
Wechsler Adult Intelligence Scale-Revised
Rorschach Test
Weschler's Memory Test

Explanation:

Case 1: This woman is probably suffering from clinical depression, which could mimic dementia in the elderly. Mild level of cognitive impairment is possible in depression and 'don't know' responses are frequently noted during bedside cognitive tests. Becks' depression inventory could be used to screen her for depression.

NART - National adult reading test: This test taps on previous word knowledge before becoming ill - hence used to estimate premorbid IQ.

The Wisconsin Card Sorting Test (WCST) is useful to test frontal executive function such as organizing and planning. It contains stimulus cards of different colour, form, and number. These are

presented to patients to sort into groups according to a single principle (e.g., to sort by colour, ignoring form and number). Persons with damage to the frontal lobes or the caudate and some persons with schizophrenia give abnormal responses.

The Wechsler Adult Intelligence Scale (WAIS-R): Most widely used intelligence test in clinical practice.The latest revision, the WAIS-III, is designed for persons 16 to 89 years of age. Wechsler Intelligence Scale for Children-III [WISC-III] is used for <16.The WAIS is composed of 11 subtests made up of six verbal subtests and five performance subtests, which yield a verbal IQ, a performance IQ, and a combined or full-scale IQ.

The correct answer is: A 55-year-old lady scores a 26/30 on the MMSE. But she responds saying, "I don't know", "I can't be bothered" for many items in the test. → Beck Depression Inventory, The family of a 60-year-old lady with traumatic brain injury is seeking compensation for the cognitive impairment sustained following RTA. You are asked to provide a report on possible premorbid IQ of this lady. → National Adult Reading Test, A 60-year-old man has poor attention span with significant difficulty organizing and planning even simple daytime activities; he vehemently denies any impairment in memory. → Wisconsin Card Sorting Test, A 16-year-old girl is disruptive and aggressive at school. She often loses her pocket money and performs poorly at school. Her brother has a moderate learning disability. → Wechsler Adult Intelligence Scale-Revised

Question **21**

Not answered

Choose...
Astereognosia
Simultagnosia
Hemisomatognosia
Anosognosia
Constructional apraxia
Prosopagnosia
Ideomotor apraxia
Bucco-facial apraxia
Ideational apraxia
Dressing apraxia

Agnosias & Apraxias

Find one term each to describe the following presentation;

70-year-old electrician suffered a left hemisphere CVA but has recovered the right arm and leg function completely. He found that although he suffered no residual weakness or loss of sensation, he could not use his tools and instruments.	Choose...
A patient finds it difficult to perform sequential actions, even though he understands and carries out simple imitations	Choose...
An inability to copy intersecting pentagons on mini-mental state examination	Choose...
A 76-year-old man with stroke has difficulty in whistling and coughing on command	Choose...
A 63-year-old man with history of younger onset dementia has difficulty to put on clothing that has been turned inside-out	Choose...

Explanation:

Ideomotor apraxia is a disorder of goal-directed movement. The patient knows what to do but not how to do it. Patients show impairment in the ability to use tools. Abnormalities include (i) abnormal amplitude of motor actions; (ii) body-part-as-object substitution, e.g. the patient uses his finger to represent a toothbrush when asked to brush his teeth; (iii) abnormal orientation of body part performing the action, and (iv) temporal errors. Improves on imitation and with the use of an actual tool. Tool use more affected than gestures.

Ideational apraxia is the inability to perform a multiple-step task. The patient does not know what to do. They exhibit an impairment in carrying out the correct sequences of actions that require ordered use of various objects and tools (Zadikoff and Lang, 2005).

Constructional apraxia is inability to construct elements into a meaningful whole. e.g., inability to draw or copy simple diagrams or figures. A lesion in the right cerebral hemisphere typically causes it.

Bucco-facial apraxia is the inability to coordinate and carry out facial and lip movements such as whistling, winking, coughing, etc., on command. The subject cannot perform skilled movements

involving the lips, mouth, and tongue in the absence of paresis.

Dressing apraxia is having the patient put on clothing that has been turned inside out.

Zadikoff, C & Lang AE. (2005). Apraxia in movement disorders. Retrieved from http://brain.oxfordjournals.org/content/brain/128/7/1480.full.pdf

The correct answer is: 70-year-old electrician suffered a left hemisphere CVA but has recovered the right arm and leg function completely. He found that although he suffered no residual weakness or loss of sensation, he could not use his tools and instruments. → Ideomotor apraxia, A patient finds it difficult to perform sequential actions, even though he understands and carries out simple imitations → Ideational apraxia, An inability to copy intersecting pentagons on mini-mental state examination → Constructional apraxia, A 76-year-old man with stroke has difficulty in whistling and coughing on command → Bucco-facial apraxia, A 63-year-old man with history of younger onset dementia has difficulty to put on clothing that has been turned inside-out → Dressing apraxia

Question **22**

Not answered

Psychiatry & Endocrine Disorders

Using the clinical descriptions given below, identify the endocrine disorder.

Choose...
Addison's disease
Cushing's syndrome
Crohn's syndrome
Hyperparathyroidism
Hyperthyroidism
Pheochromocytoma
Hypothyroidism
Diabetes mellitus
Diabetes insipidus

A 35-year-old gentleman is presenting with fatigue, muscle pain, depression, increased thirst. His wife reports an insidious change in his personality — Choose...

A 42-year-old woman with paroxysmal palpitations and raised blood pressure. She was referred to psychiatrist for severe anxiety and panic attacks — Choose...

A 56-year-old gentleman presented to GP with a history of polyuria, polydipsia, tiredness and weakness. He tested positive for urinary ketones — Choose...

A 42-year-old woman complains of dry skin, tiredness & lethargy. She has bradycardia and has some cognitive difficulties. — Choose...

Explanation:

Hyperparathyroidism is caused by an enlargement of parathyroid glands leading to too much production of parathyroid hormone and increased levels of calcium in the blood. The clinical features include back pain, bone pain, blurred vision, depression, and profound change in personality, fatigue, increased thirst, increased urinary output, itchy skin, and nausea, loss of appetite and muscle pain.

Phaeochromocytoma is characterized by increased heart rate, increased blood pressure, palpitations, and anxiety often resembling a panic attack, headaches & diaphoresis.

Diabetes mellitus is characterised by polyuria, polydipsia, tiredness, weight loss, and fatigue with excessive urinary ketones when untreated.

Hypothyroidism manifests as weight gain, constipation, dislike of cold, menorrhagia, hoarse voice, depression, cognitive difficulties, dry skin, tiredness & lethargy and bradycardia.

The correct answer is: A 35-year-old gentleman is presenting with fatigue, muscle pain, depression, increased thirst. His wife reports an insidious change in his personality → Hyperparathyroidism, A 42-year-old woman with paroxysmal palpitations and raised blood pressure. She was referred to psychiatrist for severe anxiety and panic attacks → Pheochromocytoma, A 56-year-old gentleman presented to GP with a history of polyuria, polydipsia, tiredness and weakness. He tested positive for urinary ketones → Diabetes mellitus, A 42-year-old woman complains of dry skin, tiredness & lethargy. She has bradycardia and has some cognitive difficulties. → Hypothyroidism

Question **23**

Not answered

Marked out of 4.00

Choose...
Hyperparathyroidism
Diabetes insipidus
Acute porphyria
Hypothyroidism
Diabetes mellitus
Hyperthyroidism
Addison's disease
Pheochromocytoma
Cushing's syndrome

Psychiatry & Endocrine Disorders

Using the clinical descriptions given below, identify the disorder from list above

A 30-year-old woman complains of vomiting, headaches and abdominal pain and reddish colouration of urine. She is labile in her mood and often appears confused during these attacks. This usually happens after a night out, having alcohol with friends — Choose...

A 21-year-old gentleman with polyuria, polydipsia, high plasma osmolality, low urine osmolality — Choose...

A 36-year-old woman with hirsutism, weight gain and hypertension. She is also clinically depressed. — Choose...

A 37-year-old gentleman with palpitations, excessive sweating and weight loss despite increased appetite. He also suffers from anxiety symptoms. — Choose...

Explanation:

Case 1: The scenario depicts acute intermittent porphyria (AIP). It is one of the groups of disorders of haem metabolism, characterised by neurological and psychiatric manifestations without visible cutaneous markers. AIP manifests itself by "abdomen pain, neuropathies, and constipation, but, unlike most types of porphyria, patients with AIP do not have a rash". It is an autosomal dominant disorder with the presentation starting between ages 18 and 40. It is episodic in nature, and the episodes are often triggered by certain medications including estrogens, barbiturates and benzodiazepines. Diclofenac can precipitate an episode. Psychiatric manifestations include depression, anxiety, delirium and psychosis. The most important lab test is demonstrating increased urinary porphobilinogen during acute attacks. Treatment is aimed at reducing haem synthesis by administering haemin.

Case 2: Diabetes insipidus results from impaired renal water resorption either due to a reduction in pituitary ADH secretion (cranial DI) or impaired renal response to ADH (Nephrogenic DI). This condition is characterised by polyuria, polydipsia, dilute urine, dehydration if not drinking. Plasma osmolality should be high and urine low.

Case 3: Cushing's syndrome is characterised by weight gain, menstrual irregularities, amenorrhoea, hirsutism, depression, muscle weakness, fatigue and fractures.

Case 4: Hyperthyroidism is characterised by weight loss despite increased appetite, irritability, heat intolerance, sweating, diarrhoea, tremor, frenetic activity, anxiety, emotional lability, psychosis, itch, and oligomenorrhoea.

Excerpts retrieved from: Acute Intermittent Porphyria - Medscape; http://emedicine.medscape.com/article/205220-overview (accessed April 5, 2015).

The correct answer is: A 30-year-old woman complains of vomiting, headaches and abdominal pain and reddish colouration of urine. She is labile in her mood and often appears confused during these attacks. This usually happens after a night out, having alcohol with friends → Acute porphyria, A 21-year-old gentleman with polyuria, polydipsia, high plasma osmolality, low urine osmolality → Diabetes insipidus, A 36-year-old woman with hirsutism, weight gain and hypertension. She is also clinically depressed. → Cushing's syndrome, A 37-year-old gentleman with palpitations, excessive sweating and weight loss despite increased appetite. He also suffers from anxiety symptoms. → Hyperthyroidism

Question **24**

Not answered

Marked out of

Choose...
Computed tomography
Functional MRI
Magnetic resonance spectroscopy
Magnetoencephalography
MRI venography
Single photon emission computed tomography
Magnetic resonance imaging
Positron emission tomography
MRI angiography
Contrast CT scan

Neuroimaging techniques

From the listed options choose the best answer for each of the following descriptions:

Dynamic measurement of resonance of hydrogen and phosphorus ions	Choose...
Measurement of static magnetic activity of brain	Choose...
Measurement of regional brain metabolism by tagging organic compounds in metabolic pathway	Choose...
Measurement of receptor binding and distribution using radiolabelled receptor agonists or antagonists	Choose...
Measurement of task-related changes in blood oxygenation levels	Choose...

Explanation:

Magnetic resonance spectroscopy measures the resonance-induced signal from several biologically important nuclei with an odd number of protons.

Changes in the magnetic field potentials due to neuronal activity in the brain can be captured using magnetoencephalography.

A PET scan gives direct information about neuronal metabolism.

SPECT uses radioactive compounds to study regional differences in cerebral blood flow within the brain. This records the pattern of photon emission from the bloodstream according to the level of perfusion in different regions of the brain. SPECT uses compounds labeled with single photon-emitting isotopes: iodine-123, technetium-99m, and xenon-133.

f-MRI measures task-related changes in blood oxygenation levels. Neuronal activity is associated with a local increase in blood flow, which can be observed as an increase local hemoglobin concentration. This change can be detected using a T2 sequence, called Blood Oxygen Level Dependent (BOLD) technique in fMRI. The method detects tissue perfusion rather than neuronal metabolism directly.

Ref: Section 3.3 - Neuroimaging - Kaplan and Sadocks Synopsis of Psychiatry. Excerpt retrieved from http://flylib.com/books/en/4.518.1.10/1/ (accessed April 21, 2015).

The correct answer is: Dynamic measurement of resonance of hydrogen and phosphorus ions → Magnetic resonance spectroscopy, Measurement of static magnetic activity of brain → Magnetoencephalography, Measurement of regional brain metabolism by tagging organic compounds in metabolic pathway → Positron emission tomography, Measurement of receptor binding and distribution using radiolabelled receptor agonists or antagonists → Single photon emission computed tomography, Measurement of task-related changes in blood oxygenation levels → Functional MRI

Question **25**

Not answered

Marked out of 5.00

Flag question

Frontal lobe tests

A 55-year-old man presents with cognitive problems, personality changes and disinhibition. You are performing various frontal lobe tests and the medical students on clinical attachment are keen to know the names of these tests. With the descriptions given below, identify the type of frontal lobe test:

'What is common between an apple and a banana?'	Choose...

Choose...
Proverb interpretation test
Glabellar tap
Luria motor test
Go-No Go test
Cognitive estimates testing
Lexical fluency
Trail making test
Inferential reasoning
Error correction

How tall is an average English woman? Choose...

Could you tell me as many as words as possible starting with the letter F in one minute but not names of people or places? Choose...

This test has two parts. In test A, a simple number sequence is used to join the dots. Test B uses alternating numbers and letters. Choose...

You instruct the patient to tap once in response to a single tap and to withhold a response for two taps. Choose...

Explanation:

To assess the faculty of inferential reasoning, we can use questions about the similarity between two conceptually similar objects. When testing similarity it is best to first test simple pairs such as "apples and oranges" or "desk and chair", followed by more abstract pairs such as "honesty and sincerity" or "music and melody".

To test phonological (letter) fluency, patients are asked to produce as many words as possible starting with the alphabets F, A, and S, one after the other. To test semantic (category) fluency, patients are asked to produce as many words as possible in the category of animals (or supermarket goods).

The cognitive estimates test is an attempt to prompt improbable responses that are sometimes seen in patients with frontal lesions. The questions include checking for facts such as the height of Big Ben, the population of UK, or the speed of a typical train.

Trail making test consists of two parts. In part A, a simple number sequence is used to join the dots. In part B alternating numbers and letters are used as this is more sensitive to frontal lobe dysfunction. Trail making tests are not specific to frontal functions, but also test visuomotor tracing, attention, conceptualisation and set shifting.

Impulsivity reflects a failure of response inhibition, and is seen in inferior frontal pathology. It can be assessed using the Go-No-Go task. The examiner instructs the patient to tap once in response to a single tap and to withhold a response for two taps. This test can be made more difficult by changing the initial rule after several trials (for example, "tap once when I tap twice, and not at all when I tap once").

The correct answer is: 'What is common between an apple and a banana?' → Inferential reasoning, How tall is an average English woman? → Cognitive estimates testing, Could you tell me as many as words as possible starting with the letter F in one minute but not names of people or places? → Lexical fluency, This test has two parts. In test A, a simple number sequence is used to join the dots. Test B uses alternating numbers and letters. → Trail making test, You instruct the patient to tap once in response to a single tap and to withhold a response for two taps. → Go-No Go test

Question **26**

Not answered

Marked out of 6.00

Flag question

Neuroimaging tests in psychiatry

Identify the most likely neuroimaging findings for the following psychiatric conditions

A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to ventricular enlargement, Choose...

what other feature can be seen?

A 29-year-old woman with low mood, anhedonia and poor sleep. In addition to white matter hyperintensities, what other feature can be seen?

A 77-year-old man with history of short-term memory loss and poor concentration. In addition to medial temporal atrophy, what other feature can be seen?

Choose...
Increased frontal lobe volumes
Progressive increase in brain volume in first few years of diagnosis
Poor activation of dorsolateral prefrontal cortex in executive tasks during f-MRI
Ventricular enlargement
Loss of white matter in the dorsolateral prefrontal cortex
Decreased metabolism in prefrontal and anterior cingulate cortex
Increased blood flow in parietotemporal areas in SPECT
Lower prefrontal metabolism

Choose...

Choose...

Explanation:

Case 1: The diagnosis is schizophrenia. Neuroimaging findings in schizophrenia include ventricular enlargement and loss of grey matter - especially insula, anterior cingulate cortex, medial temporal lobe and dorsolateral pre-frontal cortex with some progressive loss of brain volume in the first few years of diagnosis. fMRI shows poor dorsolateral pre-frontal cortex activation in executive tasks, while decreased N-acetyl aspartate in prefrontal cortex is seen in spectroscopy.

Case 2: The diagnosis is depressive disorder. Neuroimaging findings in depression include periventricular and deep WM hyperintensities, subcortical thalamic and striatal hyperintensities, decreased frontal and basal ganglia volumes, and decreased metabolism in prefrontal cortex.

Case 3: The diagnosis is Alzheimer's dementia. Neuroimaging findings in Alzheimer's disease include ventricular enlargement, reduced medial temporal lobe volume - especially hippocampus and a reduction in parieto-temporal fMRI activation and SPECT blood flow.

The correct answer is: A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to ventricular enlargement, what other feature can be seen? → Poor activation of dorsolateral prefrontal cortex in executive tasks during f-MRI, A 29-year-old woman with low mood, anhedonia and poor sleep. In addition to white matter hyperintensities, what other feature can be seen? → Decreased metabolism in prefrontal and anterior cingulate cortex, A 77-year-old man with history of short-term memory loss and poor concentration. In addition to medial temporal atrophy, what other feature can be seen? → Ventricular enlargement

Question **27**

Not answered

Marked out of 4.00

Flag question

Parietal lobe function

Find one term each to describe the following presentation;

A 65-year-old gentleman is unable to identify familiar objects such as key and coins by touching while his eyes are closed. He has difficulty in describe their shape and texture. However, he identifies them when shown to him.

Choose...

Choose...
Finger agnosia
Visuospatial agnosia
Agraphia
Astereognosis
Dyscalculia
Two point discrimination
Constructional apraxia
Agraphaesthesia
Right to left disorientation
Dressing apraxia

A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing — Choose...

A 73-year-old man with advanced Alzheimer's disease is unable to name fingers in his hands — Choose...

A 66-year-old woman with vascular dementia has difficulty in subtracting three from 100. — Choose...

Explanation: Astereognosis is defined as inability to recognize objects by palpation, and without visual inspection. Agraphaesthesia is defined as inability to recognize what number or alphabet is scratched on one's skin without seeing. Finger Agnosia is inability to name/number fingers. Dyscalculia is an inability to calculate.

The correct answer is: A 65-year-old gentleman is unable to identify familiar objects such as key and coins by touching while his eyes are closed. He has difficulty in describe their shape and texture. However, he identifies them when shown to him. → Astereognosis, A 67-year-old woman was unable to recognize what number is scratched into her skin without seeing → Agraphaesthesia, A 73-year-old man with advanced Alzheimer's disease is unable to name fingers in his hands → Finger agnosia, A 66-year-old woman with vascular dementia has difficulty in subtracting three from 100. → Dyscalculia

Question **28**

Not answered

Marked out of 6.00

Neurocognitive examination findings

Identify characteristic findings for each of the following situations;

Choose...
Verbal Memory Deficits
Ataxic gait
Pure Word Deafness
Homonymous hemianopia
Impaired set shifting
Stereognosis
Finger Agnosia
Dyscalculia

A 70-year-old man is diagnosed with frontal meningioma on the dominant side. He has impaired verbal fluency. What other feature can be expected? — Choose...

A 59-year-old man suffers from a left parietal stroke. He has dyscalculia. What other feature can be expected? — Choose...

A 60-year-old lady sustains a fall and knocks the back of her head against the floor. She is diagnosed with a large haematoma in the occipital region. — Choose...

Explanation: Impaired verbal fluency and impaired set shifting can be seen in frontal lobe lesions.A dominant parietal lesion can cause finger agnosia and dyscalculia. Stereognosis is normal; astereognosis can occur in parietal lesions.Cortical blindness and elementary visual hallucinations in the form of photopsia are typical in occipital infarcts.

The correct answer is: A 70-year-old man is diagnosed with frontal meningioma on the dominant side. He has impaired verbal fluency. What other feature can be expected? → Impaired set shifting, A 59-year-old man suffers from a left parietal stroke. He has dyscalculia. What other feature can be expected? → Finger Agnosia, A 60-year-old lady sustains a fall and knocks the back of her head against the floor. She is diagnosed with a large haematoma in the occipital region. → Homonymous hemianopia

Question **29**

Not answered

Marked out of 4.00

Neuropsychological measurement

Identify the neuropsychological assessment, which could be used in each of the following situations

You want to use an IQ test that has cultural and

Choose...
National Adult Reading Test
Cognitive estimates test
Mini-Mental State Examination
Stroop test
Ravens progressive matrices
Rivermead Behavioural Memory test
Trail making test
Stanford-Binet test
Luria-Nebraska battery
Digit span

lingual sensitivity in order to measure IQ of a refugee who speaks very limited English. Choose...

You want to use a tool to screen for the presence of a significant brain injury before undertaking a detailed neuropsychological examination. Choose...

You want to measure the impairment of everyday memory functioning to facilitate rehabilitation of individuals with acquired brain damage. Choose...

Explanation: Ravens Progressive Matrices test IQ using non-verbal, performance measures. It is not as culture-biased as the WAIS and does not require a fluent use of English language. Both Luria-Nebraska battery and Halstead-Reitan battery are used as screening measures to detect the presence of brain damage and localise possible areas damaged using neuropsychological tests before further detailed exploration. The Rivermead Behavioural Memory Test is designed to measure the demand made on memory by normal daily life to aid rehabilitation. It is also sensitive to change. It is not based on any theoretical model of memory.

The correct answer is: You want to use an IQ test that has cultural and lingual sensitivity in order to measure IQ of a refugee who speaks very limited English. → Ravens progressive matrices, You want to use a tool to screen for the presence of a significant brain injury before undertaking a detailed neuropsychological examination. → Luria-Nebraska battery, You want to measure the impairment of everyday memory functioning to facilitate rehabilitation of individuals with acquired brain damage. → Rivermead Behavioural Memory test

Question **30**

Not answered

Marked out of 5.00

Flag question

Choose...
Fast beta activity , Normal EEG
Posteriorly reduced alpha.
Sleep spindles
Spike and wave activity
Reduced & slow alpha waves, Flattened trace
Focal frontal slowing of EEG
Increased delta waves
Triphasic sharp waves.
Periodic K Complexes

EEG abnormalities

For each of the following scenarios choose the most likely EEG abnormality.

A 45-year-old gentleman presents with depression and anxiety with cognitive impairment. His grandfather committed suicide at the age of 65 and his father developed 'some dementia' when he was 60. Choose...

A 68-year-old man has been dependent on alcohol since the death of his wife three years ago. He develops a chest infection and gets admitted at a hospital where he develops a sudden onset of visual hallucinations and intense fear. He is visibly shaking and sweating on examination. Choose...

A 32-year-old lady has been arrested for combative behaviour at a public place. She has a history of episodic aggression and has been found to be confused and amnesic following some episodes of bizarre behaviour. Choose...

Explanation:

The first scenario is suggestive of Huntington's disease. In a number of patients, it presents with subtle psychiatric disturbances before neurological symptoms can be seen. A generalized reduction of alpha activity is an early sign; later the EEG may show a flat trace.

The second scenario is one of the delirium tremens. In patients with other types of delirium, usually the EEG shows a diffuse slowing of the background rhythm. In delirium tremens, the EEG shows fast activity or can be normal.(Ref: Gleason OC. Delirium. Am Fam Physician. 2003 Mar 1;67(5):1027-1034).
The last scenario is suggestive of complex partial seizures. Spike and wave activity is classical of epilepsy; in this case it may initially be lateralized.

The correct answer is: A 45-year-old gentleman presents with depression and anxiety with cognitive impairment. His grandfather committed suicide at the age of 65 and his father developed 'some dementia' when he was 60. → Reduced & slow alpha waves, Flattened trace, A 68-year-old man has been dependent on alcohol since the death of his wife three years ago. He develops a chest infection and gets admitted at a hospital where he develops a sudden onset of visual hallucinations and intense fear. He is visibly shaking and sweating on examination. → Fast beta activity , Normal EEG, A 32-year-old lady has been arrested for combative behaviour at a public place. She has a history of episodic aggression and has been found to be confused and amnesic following some episodes of bizarre behaviour. → Spike and wave activity

Question **31**

Not answered

Marked out of 5.00

Flag question

Behavioural genetics

Choose the most relevant pattern of inheritance for each scenario described below.

A 32-year-old teacher gives birth to a female child. The child was hypotonic at birth and needed extended resuscitation after birth. Umbilical hernia and Simian crease were noted.

Choose...

Chromosomal translocation, Meiotic nondisjunction
Uniparental disomy, Parent of origin effect
X-linked recessive
Autosomal recessive
Polymorphism
Trinucleotide repeat syndrome
Autosomal dominant
X-linked dominant

An 18 months old female child presents with developmental stagnation, deceleration of the head circumference, and psychomotor regression with crying spells. Odd circular movements around the midline of the body are also noted. Later, autistic traits become apparent.

Choose...

A 4 years old child shows severe mental retardation, seizures, and ataxic gait. There is no family history of similar illness. Facial features reveal blue eyes with the flattened occiput, prominent jaw, wide mouth and pointed chin.

Choose...

Explanation: Down Syndrome - Trisomy 21 (95% of Down syndrome) is caused by a meiotic nondisjunction event (in 88% of trisomies)or extra chromosome 21 material (from chromosome 14) due to a Robertsonian translocation in the karyotype of one of the parents.

More than 99 percent of classic Rett syndrome cases result from new mutations in the MECP2 gene, but a few families have been described with an X-linked dominant pattern of inheritance.

Angelman syndrome is caused by the loss of the normal maternal contribution to a region of chromosome 15, most commonly by deletion, uniparental disomy, translocation, or single gene mutation in that region.

The correct answer is: A 32-year-old teacher gives birth to a female child. The child was hypotonic at birth and needed extended resuscitation after birth. Umbilical hernia and Simian crease were noted. → Chromosomal translocation, Meiotic nondisjunction, An 18 months old female child presents with developmental stagnation, deceleration of the head circumference, and psychomotor regression with crying spells. Odd circular movements around the midline of the body are also noted. Later, autistic traits become apparent. → X-linked dominant, A 4 years old child shows severe mental retardation, seizures, and ataxic gait. There is no family history of similar illness. Facial features reveal blue eyes with the flattened occiput, prominent jaw, wide mouth and pointed chin. → Uniparental disomy, Parent of origin effect

Question **32**

Not answered

Marked out of 4.00

Choose...
Noradrenaline
Dopamine
Neuropeptide Y
Glycine
Endorphin
Glutamate
GABA
Histamine
Acetylcholine
Anandamide
Nitric oxide
Serotonin
Adrenaline

Neurochemical mediators

For each of the following description, choose the most appropriate neurotransmitter from the list above.

Acts predominantly as a hormone; the immediate precursor is a major neurotransmitter. — Choose...

The major inhibitory neurotransmitter of the central nervous system in humans. — Choose...

Inhibition of reuptake of this transmitter can produce intense reward and sense of satisfaction in cocaine users. — Choose...

Substances present in tobacco smoke mimic the action of this neurotransmitter. — Choose...

Epinephrine (also known as adrenaline) is a hormone and a neurotransmitter. Tyrosine is first oxidized to L-DOPA, which is subsequently decarboxylated to give dopamine, oxidation of which gives norepinephrine, which is methylated to give epinephrine. [Kaplan & Sadock's Synopsis of Psychiatry: Behavioral Sciences/Clinical Psychiatry, 10th Edition Pg 103]

The two major amino acid neurotransmitters are GABA and glutamate. GABA is an inhibitory amino acid, and glutamate is an excitatory amino acid.[Kaplan & Sadock's Synopsis of Psychiatry: Behavioral Sciences/Clinical Psychiatry, 10th Edition Pg 108].

Amphetamines cause the release of dopamine and cocaine blocks the uptake of dopamine in the nucleus accumbens. It is thought that cocaine and amphetamine exert the hedonistic effect due to increased dopaminergic neurotransmission in Nucleus accumbens.[Kaplan & Sadock's Synopsis of Psychiatry: Behavioral Sciences/Clinical Psychiatry, 10th Edition Pg 104].

The psychoactive component of tobacco is nicotine, which affects the central nervous system (CNS) by acting as an agonist at the nicotinic subtype of acetylcholine receptors. [Kaplan & Sadock's Synopsis of Psychiatry: Behavioral Sciences/Clinical Psychiatry, 10th Edition Pg 440]

The correct answer is: Acts predominantly as a hormone; the immediate precursor is a major neurotransmitter. → Adrenaline, The major inhibitory neurotransmitter of the central nervous system in humans. → GABA, Inhibition of reuptake of this transmitter can produce intense reward and sense of satisfaction in cocaine users. → Dopamine, Substances present in tobacco smoke mimic the action of this neurotransmitter. → Acetylcholine

Question

33

Not answered

Marked out of

Neural tube derivatives

Match the brain components with their derivatives of the neural tube

Caudal part of medulla oblongata	Choose...
Thalamus	Choose...
Corpus striatum	Choose...
Hypothalamus	Choose...
Red nucleus	Choose...

Choose...
Metencephalon
Myelencephalon
Telencephalon
Diencephalon
Mesencephalon
Rhomboncephalon
Paraencephalon
Rosterencephalon

Explanation: During embryonic development the midline neural tube differentiates into the following vesicles.

- **Prosencephalon** which differentiates into the Telencephalon (cerebrum, striatum and pallidum) and Diencephalon (thalamus, subthalamus, pineal body, hypothalamus and epithalamus).
- **Mesencephalon** (midbrain) has two parts - tectum or corpora quadrigemina, made up of the superior & inferior colliculi and tegmentum containing the red nucleus and periaqueductal grey matter.
- **Rhombencephalon** which differentiates into Metencephalon that forms pons, the rostral part of the medulla oblongata and cerebellum and Myelencephalon that forms the caudal part of medulla oblongata.

The correct answer is: Caudal part of medulla oblongata → Myelencephalon, Thalamus → Diencephalon, Corpus striatum → Telencephalon, Hypothalamus → Diencephalon, Red nucleus → Mesencephalon

Question **34**

Not answered

Marked out of 5.00

EEG findings

You are working in the EEG clinic. You are requested to correlate some EEG findings with probable diagnoses associated with them. Identify the disorders for each of the following EEG findings

Episodic discharges every 1-3 seconds with variable focal waves.	Choose...
Generalised polyspike and wave pattern	Choose...
Triphasic wave pattern	Choose...
Focal slow wave activity	Choose...
1 to 2 Hz periodic generalised sharp waves against a low and slow background	Choose...

Choose...
Infantile spasms
Depressive pseudodementia
Angelman's syndrome
Huntington's chorea
Myoclonic epilepsy
Herpes simplex encephalitis
Hepatic encephalopathy
Structural lesions
CJD
Neurosyphilis

Explanation:

CJD- Generalized periodic 1-2 Hz sharp waves are seen in nearly 90% patients with sporadic CJD. Less often in familial / hormonal transplant-related forms. It is not seen in the variant form.

Myoclonic epilepsy- Generalized multiple spike-wave discharges.

Diffuse generalized slowing is seen in metabolic and endocrine disorders.

Triphasic waves: 1.5 to 3.0 per second high-voltage slow-waves especially in hepatic encephalopathy. There is focal slowing or focal spike activity in structural lesions of the brain.

In herpes simplex encephalitis, there are episodic discharges every 1-3 seconds with variable focal waves over the temporal lobes.

The correct answer is: Episodic discharges every 1-3 seconds with variable focal waves. → Herpes simplex encephalitis, Generalised polyspike and wave pattern → Myoclonic epilepsy, Triphasic wave pattern → Hepatic encephalopathy, Focal slow wave activity → Structural lesions, 1 to 2 Hz periodic generalised sharp waves against a low and slow background → CJD

Question **35**

Not answered

Marked out of 5.00

Synthetic enzymes of neurotransmitters

Identify the enzyme involved in the immediate synthesis of each of the following neurotransmitters

Dopamine	Choose...
Norepinephrine	Choose...
Serotonin	Choose...
Acetylcholine	Choose...
GABA	Choose...

Choose...
5-hydroxyindole acetic acid
Homovanillic acid
Tryptophan hydroxylase
GABA transaminase
Dopamine beta-hydroxylase
Glutamic acid decarboxylase
Choline acetyltransferase
Phenylethanolamine-N-methyltransferase
Acetylcholinesterase
Dopa decarboxylase

Explanation:

- DOPA decarboxylase produces dopamine from dopa.
- Dopamine-b-hydroxylase modulates norepinephrine production from dopamine.
- Phenylethanolamine-N-methyltransferase modulates conversion of norepinephrine to epinephrine.
- The synthetic enzyme for acetylcholine is choline acetyltransferase (CAT) and not acetylcholinesterase; the latter is the breakdown enzyme.
- Glutamic acid decarboxylase (GAD) produces GABA from glutamate.

The correct answer is: Dopamine → Dopa decarboxylase, Norepinephrine → Dopamine beta-hydroxylase, Serotonin → Tryptophan hydroxylase, Acetylcholine → Choline acetyltransferase, GABA → Glutamic acid decarboxylase

Finish review

PREVIOUS ACTIVITY
EMIs Neuroscience Part 1

NEXT ACTIVITY
Advanced Neuroscience *NEW* December 2018

Jump to...

Drop here!

© 2022 SPMM Course Limited, UK.

Refund Policy | Terms & Conditions